

COMPREHENSIVE ITC327W STAKEHOLDER PROJECT REPORT

Project: Online Student Application and Admission Management System

Target Stakeholder: Setshabelo Primary School

Student Name: Lonwabo Sifumba

Student Number: 220049798

Reporting Phase: Unit 2 / Phase 1

Date: 31 August 2026

SECTION 1: STAKEHOLDER REQUIREMENTS AND NEEDS ASSESSMENT

1. Stakeholder Information

- **Stakeholder:** Setshabelo Primary School
- **Primary Users:** Parents/Guardians and School Administration/Teachers
- **System Type:** Online Student Application and Admission Management System

2. Stakeholder Problem

Setshabelo Primary School experiences a high intake of student applications. Currently, the application and admission process is mainly paper-based, creating several operational challenges:

- Parents must complete physical application forms.
- School staff must manually collect and process applications.
- Paper documents can be lost, damaged, or incorrectly filed.
- Processing a large volume of applications takes considerable time.
- Teachers or administrators struggle to organize applicant information efficiently.
- Tracking application progress is difficult for both staff and parents.
- Parents must visit the school physically to submit applications or inquire about status updates.

3. Stakeholder Needs Assessment

The stakeholder requires a digital system to modernize and streamline the application and admission process.

Stakeholder Need	Description
Online Applications	Parents can apply for their children online.
Parent Registration	Parents can create a secure account.
Student Info Management	Parents can provide all required learner information.
Document Management	Supporting documents can be submitted digitally.
Application Tracking	Parents can check application progress or status online.
Administration System	Dedicated access for school staff to administrative functions.
Application Management	Teachers/administrators can view and manage applicant records.
Reduced Paperwork	Minimize reliance on physical paper forms.
Secure Information	Protect sensitive student and parent data.
Easy Access	User-friendly interfaces tailored for both parents and school staff.

4. Stakeholders and Detailed Requirements

A. Parents / Guardians

Parents will primarily use the front-facing user application to:

- Register an account and log in securely.

- Enter personal contact details.
- Register/add learner information.
- Complete and submit online admission applications.
- Upload required supporting documents.
- View application status and receive progress notifications.
- Update permitted details when necessary.

B. School Administration and Teachers

Staff will use the administrative panel to:

- Log into a secure administrative portal.
- View, search, and review submitted student applications.
- Verify applicant details and supporting documents.
- Update application statuses (e.g., Approve, Reject, Pending Verification).

- Manage student and parent records based on authorized access levels.
- Generate and export relevant administrative reports.

5. Functional Requirements

Parent / Guardian Functions

- **FR01:** The system shall allow parents to register an account.
- **FR02:** The system shall allow parents to log in securely.
- **FR03:** The system shall allow parents to register their children's information.
- **FR04:** The system shall allow parents to complete an online admission application.
- **FR05:** The system shall allow parents to submit an application electronically.
- **FR06:** The system shall store submitted application information securely.
- **FR07:** The system shall allow parents to view their application status.

- **FR08:** The system may allow parents to upload required supporting documents.
- **FR09:** The system shall allow parents to log out securely.

School Administration Functions

- **FR10:** The system shall allow authorized staff to log into the administration system.
- **FR11:** The system shall display submitted applications to authorized staff.
- **FR12:** The system shall allow staff to search for applications.
- **FR13:** The system shall allow authorized staff to view applicant details.
- **FR14:** The system shall allow authorized staff to update application status.
- **FR15:** The system shall allow staff to manage student application records.
- **FR16:** The system shall provide different access levels based on user roles.
- **FR17:** The system shall allow authorized staff to manage administrative information.

6. Proposed User Roles

1. **Parent / Guardian:** Register account, manage learner info, submit applications, upload documents, track status.
2. **Teacher / Administrator:** Access admin portal, review applications, verify documents, update status, view student records.
3. **System Administrator:** Manage user roles and system permissions, maintain platform infrastructure.

7. Non-Functional Requirements

- **Usability:** Simple layout suitable for users with varying technical skills; clear instructions.
- **Security:** Secure authentication, encrypted database storage, strict role-based authorization, protected credentials and API keys.
- **Performance:** Efficient database querying with fast application submission processing and multi-user handling.
- **Reliability:** Accurate data persistence minimizing risks of lost records or corrupted data.

- **Accessibility:** Responsive web/mobile cross-browser support.

8. Technology Requirements & Architecture

- **Mobile / Frontend (Flutter):** Provides mobile/web interface for parent registration, application submission, status tracking, and admin capabilities.
- **Backend API (ASP.NET Core):** Handles business logic, security middleware, application processing, and data routing.
- **Database & Auth (Supabase):** Manages user authentication, database persistence, role management, and file storage for supporting documents.

9. Data Requirements

- **Parent Information:** Full name, contact details, email address, physical address.
- **Learner Information:** Full name, date of birth, identity number, grade applying for, previous school history.

10. Priority Requirements

- **High Priority (Must Have):** Parent registration, secure login, application creation/submission, admin portal access, application review, status management, secure database storage.
- **Medium Priority (Should Have):** Application status tracking updates, document upload capabilities, application search and filter tools.
- **Low Priority (Could Have):** Automated batch messaging/notifications, advanced analytical reporting.

SECTION 2: HANDWRITTEN MEETING & SPECIFICATION NOTES (24/08/26)

Meeting Agenda & Discussion Points

- **Date:** 24 August 2026
- **Location/Context:** Setshabelo Primary School Preliminary Planning

Key Items Discussed:

1. **Attendance Tracking:** Integration possibilities for student/staff attendance tracking.
2. **Registration & Admission Workflow:**
 - Application tracking features for parents.

- Simplification of the registration/application forms.
- Document acceptance (supporting uploads).
- **Admission Committee Review:** Document verification flow involving Principal, Clerks, and Committee members.

Platform-Specific Functionality Breakdown:

- **Mobile App Features:**

- User Registration
- Information Updates
- Application Tracking & Status Updates

- **Website Features:**

- School Achievements & Announcements
- Information Updates
- Public Registration Portal

- **Shared Features (App & Website):**

- **Bulk SMS:** Automated notification dispatch for application updates and attendance.
- **Phone Scanning:** QR/Bar code scanning features for registration and attendance verification.

SECTION 3: PERSONAL PROJECT PROGRESS REPORT & FEASIBILITY ANALYSIS

1. Purpose of Contribution

Assigned responsibility involved conducting a complete feasibility study evaluating whether the proposed system could realistically be completed within the available semester using available tools, resources, skills, and time. The evaluation covered technical, operational, economic, schedule, legal, and ethical dimensions.

2. Detailed Feasibility Findings

Feasibility Area	Decision	Summary & Key Reasons
Technical	Feasible with changes	Core stack (Flutter, ASP.NET Core, Supabase) is available. Risks include backend-frontend integration, technical skill variances, and

Feasibility Area	Decision	Summary & Key Reasons
		complex features (e.g., scanning, batch messaging).
Operational	Feasible with changes	System addresses manual paperwork challenges, but must remain intuitive for parents and administrative staff across various device types.
Economic	Feasible	Focuses on resource access rather than commercial finance. Required hardware, internet, software, and Supabase tiers are available.
Schedule	Feasible with changes	Tight semester timeline. Scope expansion (such as batch messaging) must be carefully controlled to complete core modules on time.
Legal	Feasible with changes	Sensitive personal identification, passwords, and API keys must be kept out of public GitHub repositories and secured.
Ethical	Feasible with changes	Requires responsible data collection, strict user privacy, and continuous alignment with stakeholder expectations.

3. Identified Risks and Challenges

1. Limited semester timeframe for full end-to-end development, integration, and testing.
2. Skill gaps in advanced Flutter, ASP.NET Core, or Supabase usage.
3. Integration bottlenecks between Flutter client app, ASP.NET Core API, and Supabase database.

4. Scope expansion caused by additional stakeholder feature requests (e.g., batch SMS, phone scanning).
5. Delay of core features due to premature focus on secondary/complex functions.
6. Data privacy and security risks if credential leaks occur.

4. Group Recommendations

- **Prioritize Core Requirements:** Complete high-priority admission and registration functions prior to attempting optional features.
- **Schedule Discipline:** Follow and continuously update the project schedule (Microsoft Project).
- **Early Integration Testing:** Test communication between Flutter, ASP.NET Core, and Supabase early in development.
- **Credential Protection:** Enforce strict policy against committing passwords, environment secrets, and API keys to public repositories.
- **Scope Management:** Postpone lower-priority functions (e.g., bulk SMS, barcode scanning) if they threaten scheduled milestones.

5. Conclusion & Personal Reflection

The proposed system for Setshabelo Primary School is **feasible with changes**. While technical foundations are strong, success requires scope control, task prioritization, and early integration testing. Completing this assessment highlighted the necessity of balancing technical capability against project constraints like time, privacy, and skill balance.

Report Prepared By: Lonwabo Sifumba

Student Number: 220049798

Date: August 2026

24/08/26

* LMS

- Removed due to Financial Restrictions.

* Student & Card Sumner.

- "Paused" - Financial Restrictions.

* Store.

- Cancelled $\Rightarrow$ Provided by supplier.

* Bulk SMS

* Centralized System. Website

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